

Σ - Λ Theory and Cortex

A Formal Walkthrough of Resource-Bounded Articulation, Predictive Memory, and Local Corrigibility

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refine on certified consequential witnesses;
coarsen only when redundancy is separately certified.

Purpose. This whitepaper presents the present mathematical core of the Sigma–Lambda research program and its executable Cortex realization after a formal audit. It is intentionally modular: several components are re-derivations or specializations of established results, several are project-specific synthesis layers, and there is no theorem proving the entire architecture end to end. The paper therefore separates (i) mathematical status, (ii) provenance/novelty, and (iii) empirical validation. The central scientific object is a resource-bounded process for discovering, maintaining, revising, and composing consequential hidden structure in grounded information streams.

Contents

Abstract	2
1 Scientific Identity and Scope	2
1.1 Substrate and articulation	2
1.2 What is grounded and what is discovered	2
1.3 Dependency architecture, not an end-to-end theorem	3
1.4 Status and novelty are separate axes	3
2 Articulation, Resolution, and Continuation Laws	4
2.1 Resolution-indexed articulated state	4
2.2 Continuation law	4
2.3 Why the quotient matters	5
3 Dynamic Sufficiency and Minimal Stable Memory	5
3.1 Partitions and lumpability	5
3.2 Synchronous refinement	5
3.3 Structural and computational interpretations of memory	6
4 Coherent Reduction, Geometry, and Topology Across Articulations	6
4.1 Projectability of predictive quotients	6
4.2 Refinement ultrametric	6
4.3 Operational reachability topology	7
4.4 Fisher geometry	7
5 Resource-Bounded Articulation and the Discoverability Frontier	7
5.1 Exact memory can be too large	7
5.2 Witnesses and refinement	7
5.3 Synergy and witness hypergraphs	8
5.4 Blind Information Well	8
6 Persistent Entities and Transformation Invariance	8
6.1 The entity problem	9
6.2 Cycle consistency and discrete holonomy	9
6.3 Global synchronization and gauge	9
6.4 Selective nuisance subgroup	9
6.5 Structural entity proposals and predictive veto	10
6.6 Prototype refinement	10
7 Simultaneous Binding	10
7.1 Why frame-level assignment must be injective	10
7.2 Bootstrap guard	10
7.3 Resource boundary	11
8 Residuals, Reliability, and Endogenous Noise Revision	11
8.1 Residual evidence	11
8.2 Three-way residual state	11
9 Relations, Events, and Causal Evidence	11
9.1 Relation/event posteriors	11
9.2 Intervention-sensitive causal contrast	12
10 Streaming Articulation	12
10.1 Finite online state	12
10.2 Piecewise-stationary theorem interface	13

10.3 Coverage and recovery horizons	13
10.4 Anytime monitoring	13
10.5 Structural invalidation versus kernel change	14
10.6 What is proved here versus inherited	14
10.7 Restricted streaming theorem	14
10.8 Regret bound	15
11 Computation and Amortization	15
11.1 Maintenance plus discovery	16
11.2 Entropy is not the controller	16
11.3 What amortization does not solve	16
12 Cortex v0.7: Dependency-Directed Execution	16
12.1 Operational state	17
12.2 Local statistics, local re-derivation	17
12.3 Synchronization hardening	17
12.4 Entity-invariance co-refinement guard	17
12.5 Capacity as an explicit resource boundary	18
12.6 Frozen v0.7 evidence and internal baseline	18
13 Status Ledger: What Is Proved, Validated, Conditional, or Open	18
14 Known Failure Frontiers	19
14.1 Search complexity	20
14.2 Equality is harder to certify than inequality	20
14.3 Pure simplification	20
14.4 Identity fragmentation under strong corruption	20
14.5 External validity	20
15 Interpretive Summary	20
16 Roadmap	21
A Notation Reference	21
B Compact Proof Sketch: Why Stable Refinement Terminates in the Finite Case	22
C Proof Architecture of the Restricted Streaming Composition	22
D Internal Technical Provenance	23
E Selected Mathematical Context and Provenance	23

Abstract

Sigma–Lambda theory separates an inexhaustible substrate, denoted Σ , from finite or resource-bounded articulations, denoted Λ . Its current scientific form is best understood as a *modular research program*, not as one proved end-to-end theory. The finite predictive core re-derives and specializes established ideas from predictive sufficient statistics/causal states, Markov lumpability, and partition refinement. The resource-bounded branch is closely related to rate–distortion and information-bottleneck formulations. The synergy branch uses standard parity/PID phenomena as exact counterexamples to purely marginal search. Downstream entity, invariance, binding, residual, relation, and causal modules are validated finite learners whose interfaces are project-specific but whose ingredients draw on established mathematical tools.

The project-specific object under investigation is the composition of these ingredients around an articulation discipline: preserve only consequential distinctions; refine automatically when positive witnesses certify that a merge is invalid; coarsen only at separately certified recompression checkpoints; expose unresolved evidence and resource failure; and maintain learned structure locally rather than recomputing a global world model after every observation. Cortex v0.7 is the current executable realization of that discipline.

This revision makes three scope points explicit. First, the dependency arrows between modules are architectural interfaces, not a theorem that estimation error propagates benignly through the full stack. Second, theorem status and novelty are orthogonal: several “exact” results are established results re-derived in project notation. Third, the restricted streaming theorem is a composition theorem conditional on finite-state Markov closure, coverage/margin lemmas, homogeneous monitoring blocks, and dwell-time assumptions; its proof architecture is included here rather than being delegated solely to an external status label.

1 Scientific Identity and Scope

1.1 Substrate and articulation

The project begins with a scope distinction:

$$\Sigma \rightsquigarrow \Lambda.$$

Here Σ denotes substrate-excess: whatever reality or generative source exists beyond a particular representation. Λ denotes an articulation: a finite, measurable, computational, or otherwise resource-bounded representation constructed from observations.

The modern formal program does *not* require a complete ontology of Σ . The substrate distinction functions as an anti-totalization guard: no finite learned model is identified with the totality of what exists. What the mathematics directly manipulates are articulated observations and the structures induced over them.

Let the observation stream be

$$P : \Sigma \rightsquigarrow \mathcal{X}^{\mathbb{N}}, \quad \omega = (X_1, X_2, \dots).$$

At time t , the learner receives only the finite prefix $X_{1:t}$. The scientific objective is therefore:

Recover and maintain the **coarsest resource-feasible articulation** that preserves the consequential distinctions supported by current evidence.

1.2 What is grounded and what is discovered

The current theory distinguishes:

- **Grounded primitives:** sensor variables, observation order, admissible resolution/refinement operations, the chosen consequence target, a resource budget, and optionally an action/intervention channel.

- **Derived articulations:** partitions, latent states, transformations, entity hypotheses, relations, temporal motifs, quotient states, and causal hypotheses constructed from grounded observations.

Thus the correct claim is *endogenous discovery of derived structure over grounded primitive observations*, not unrestricted invention of primitive degrees of freedom.

1.3 Dependency architecture, not an end-to-end theorem

The finite predictive core is organized as

$$\begin{aligned} \Sigma &\rightsquigarrow \text{grounded articulation} \rightarrow \text{continuation laws} \\ &\rightarrow \text{predictive equivalence} \rightarrow \text{predictive quotient} \\ &\rightarrow \text{dynamic sufficiency} \rightarrow \text{minimal stable memory.} \end{aligned}$$

Downstream learning modules depend on outputs of that core:

$$\text{stable memory} \rightarrow \begin{cases} \text{resource-sensitive articulation search,} \\ \text{entity / invariance structure,} \\ \text{residual / noise modelling,} \end{cases}$$

then

entities + invariance + residual control $\rightarrow$ simultaneous binding $\rightarrow$ relations/events $\rightarrow$ interventional causal evidence.

The arrows above are dependency/interface statements. No theorem in the current program proves that finite-sample error or uncertainty in an upstream predictive quotient propagates through entity discovery, binding, residual learning, relation induction, and causal contrast while preserving an end-to-end guarantee. Each module currently has its own assumptions and validation regime. The architecture is therefore a pipeline of locally justified components whose composition is empirical/contractual except where an explicit composition theorem is stated.

Three non-identifications remain essential:

$$\begin{aligned} \text{topological persistence} &\neq \text{predictive relevance} \neq \text{entity identity,} \\ \text{unexplained residual} &\neq \text{noise,} \quad \text{predictive relation} \neq \text{causal relation.} \end{aligned}$$

1.4 Status and novelty are separate axes

The word “exact” in this paper means exact *under the stated assumptions*; it does not mean novel to Sigma-Lambda. In particular:

- predictive-equivalence classes as minimal sufficient predictive states are the causal-state construction of computational mechanics [2];
- finite dynamic stability is a lumpability/coarsest-partition problem related to classical Markov aggregation and partition-refinement algorithms [1, 3];
- resource-risk compression is structurally related to rate-distortion and information bottleneck theory [4, 5];
- parity-style pure synergy is a standard multivariate-information phenomenon, naturally described by partial-information-decomposition language [6];

- group synchronization, conjugate Beta-Bernoulli updating, and assignment are established tools used here inside a particular continual-articulation architecture.

The project-specific contribution under investigation is therefore primarily *composition*, *scope discipline*, *negative results*, *witness/recompression control*, *entity-invariance synthesis*, *restricted streaming composition*, and *dependency-directed execution*, rather than a claim that all mathematical ingredients are new.

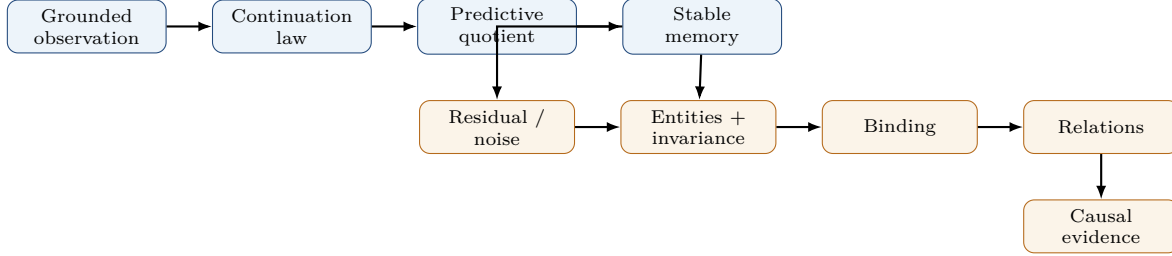


Figure 1: Dependency architecture of the current program. Arrows are interfaces; they are not an end-to-end error-propagation theorem.

2 Articulation, Resolution, and Continuation Laws

2.1 Resolution-indexed articulated state

Let $r \in \mathcal{R}$ index an admissible resolution or articulation level. Write

$$A_r : \mathcal{X} \rightarrow X_r$$

for the articulation map into a state space X_r . For two resolutions $r \preceq s$, coherence requires that the finer articulation reduce consistently to the coarser one through a map

$$\pi_{s \rightarrow r} : X_s \rightarrow X_r, \quad A_r = \pi_{s \rightarrow r} \circ A_s.$$

When these maps compose,

$$\pi_{u \rightarrow r} = \pi_{s \rightarrow r} \circ \pi_{u \rightarrow s}, \quad r \preceq s \preceq u,$$

the resolution family forms a coherent reduction system.

The resolution structure is a formal scaffold. The existence of coherent articulation maps is an axiom/interface requirement; a universal procedure that discovers the entire resolution lattice from arbitrary raw reality is not currently proved.

2.2 Continuation law

For an articulated state $x \in X_r$, define its continuation law

$$\Psi_r(x) = \mathcal{L}(\text{future consequential observations} \mid X_r = x).$$

Here $\mathcal{L}(U \mid V = v)$ denotes the conditional probability law (distribution) of the random object U given $V = v$. Thus $\Psi_r(x)$ is itself a probability measure (or probability kernel value), not a scalar loss. The future may be the full future trajectory or a task-relative consequence variable Y . The formalism is consequence-relative: changing what counts as a consequence changes the equivalence relation and can change the resulting memory.

Two states are predictively equivalent when their continuation laws agree:

Definition 2.1 (Predictive equivalence).

$$x \equiv_r y \iff \Psi_r(x) = \Psi_r(y).$$

The equivalence classes form the predictive quotient

$$Q_r = X_r / \equiv_r .$$

Established result / project re-derivation

Under standard measurable sufficiency conditions, the equivalence classes of histories/states with identical predictive future laws form the minimal sufficient predictive representation. This is the causal-state/minimal-predictive-statistic result of computational mechanics [2], re-expressed here at an arbitrary articulated resolution r . Any other sufficient representation factors through this quotient up to the usual almost-sure/isomorphism qualifications.

2.3 Why the quotient matters

If two articulated states induce the same future law, retaining their difference cannot improve prediction for the chosen consequence target. The predictive quotient therefore captures a minimality principle:

representation complexity should track consequential distinction, not raw descriptive detail.

This gives the minimality motivation for compression. Operationally, however, automatic coarsening is not inferred from absence of a witness; redundancy must be separately certified at a recompression checkpoint (Section 5.2).

3 Dynamic Sufficiency and Minimal Stable Memory

3.1 Partitions and lumpability

Let $X = \{1, \dots, m\}$ be a finite articulated carrier with transition kernel K . This section specializes classical strong lumpability/state-aggregation ideas [1] and uses a coarsest-partition refinement pattern closely related to standard partition-refinement algorithms [3]. A partition Π groups microstates into macrostate blocks. For state i , let $A_\Pi K_i$ denote the row K_i aggregated onto the blocks of Π .

Define the one-step dynamic stability distortion

$$D_{\text{dyn}}^{(1)}(\Pi; K) = \max_{B \in \Pi} \max_{i, j \in B} D_{\text{TV}}(A_\Pi K_i, A_\Pi K_j).$$

Then

$$D_{\text{dyn}}^{(1)}(\Pi; K) = 0$$

if and only if states hidden inside each block induce identical block-level transition laws. This is the finite one-step dynamic stability/lumpability condition used by the exact memory theorem.

3.2 Synchronous refinement

Start from a supplied coarse partition $\mathcal{P}_0$. Whenever two states in the same block disagree in their aggregated continuation law, split the block. Synchronous iteration yields

$$\mathcal{P}_0 \preceq \mathcal{P}_1 \preceq \dots \preceq \mathcal{P}_\star.$$

For a finite carrier, refinement terminates after finitely many strict splits.

Theorem 3.1 (Finite coarsest stable refinement; project specialization). *For a finite articulated carrier and a supplied initial coarse partition $\mathcal{P}_0$, synchronous continuation-law refinement terminates at a partition $\mathcal{P}_\star$ that is dynamically stable and is the unique coarsest dynamically stable refinement of $\mathcal{P}_0$.*

Within this finite Markov specialization, memory is therefore represented as a quotient object rather than introduced as an independent primitive:

$$\mathcal{M}_\star = \mathsf{X}/\mathcal{P}_\star.$$

3.3 Structural and computational interpretations of memory

The project gives memory two roles.

Structural role. $\mathcal{M}_\star$ is the minimal stable refinement needed to preserve consequential distinctions.

Computational role. Once a pattern is integrated, memory caches articulated structure so that future processing updates sufficient statistics instead of rediscovering structure from the entire past.

discover once $\rightarrow$ integrate $\rightarrow$ maintain cheaply $\rightarrow$ reopen only on witness.

4 Coherent Reduction, Geometry, and Topology Across Articulations

4.1 Projectability of predictive quotients

A fine predictive quotient need not deterministically collapse to a coarse predictive quotient. Suppose $r \preceq s$. A deterministic quotient transport

$$\bar{\pi}_{s \rightarrow r} : Q_s \rightarrow Q_r$$

exists only if every fine predictive class maps entirely into one coarse predictive class. If a fine class straddles multiple coarse quotient states, projectability fails and the tower of quotients is not a deterministic hierarchy.

This is an important negative result: a hierarchy of observational resolutions does not automatically induce a hierarchy of predictive states.

4.2 Refinement ultrametric

When a family of partitions is genuinely nested,

$$\mathcal{P}^{(0)} \preceq \mathcal{P}^{(1)} \preceq \dots,$$

one may define a separation depth

$$\ell(x, y) = \min\{k : x, y \text{ are separated at level } k\}$$

and an ultrametric, for example,

$$d_U(x, y) = 2^{-\ell(x, y)}.$$

The strong triangle inequality

$$d_U(x, z) \leq \max\{d_U(x, y), d_U(y, z)\}$$

follows from hierarchical refinement. The geometry therefore reflects *when* two states become consequentially distinguishable rather than their Euclidean distance.

4.3 Operational reachability topology

Given an operational closure/reachability relation, neighborhoods may be defined by states reachable through admissible refinement or continuation operations. This creates a finite/order-theoretic topology. Its role is to describe which articulations can be operationally accessed from which others; it is distinct from the topological data analysis used later for structural entity proposals.

4.4 Fisher geometry

When continuation laws form a regular parametric family $p(y \mid \theta)$, the Fisher information

$$I(\theta) = \mathbb{E}_\theta [\nabla_\theta \log p(Y \mid \theta) \nabla_\theta \log p(Y \mid \theta)^\top]$$

provides a local statistical geometry on an articulation manifold. This extension is conditional on regularity assumptions; it is not part of the minimal finite quotient theorem.

5 Resource-Bounded Articulation and the Discoverability Frontier

5.1 Exact memory can be too large

Exact predictive equivalence may yield a large or infinite quotient. The framework therefore introduces a resource-relative optimization. Let $R(\mathcal{P})$ measure predictive distortion/risk under a coherent partition or quantizer $\mathcal{P}$. Then

$$C^*(\varepsilon) = \min_{\mathcal{P}: R(\mathcal{P}) \leq \varepsilon} \text{Cost}(\mathcal{P}).$$

This is structurally a rate-distortion problem with an application-specific representation cost and distortion functional, and is closely related to the information bottleneck when the preserved quantity is predictive/relevant information [4, 5]. The Sigma-Lambda-specific restriction is that the chosen approximation must remain a coherent articulation usable by the downstream refinement/runtime machinery.

A subtlety matters: threshold closeness of continuation laws is not itself an equivalence relation because closeness need not be transitive. Approximation must therefore be represented by an actual partition/quantizer satisfying a within-block distortion constraint.

5.2 Witnesses and refinement

A *witness* is positive evidence that a currently merged distinction is consequential. In finite dynamic memory, a non-lumpability witness is a pair i, j inside one block such that

$$D_{\text{TV}}(A_{\Pi}K_i, A_{\Pi}K_j) > 0.$$

The automatic branch is therefore one-sided:

refine on certified positive consequential witnesses.

Exact or approximate coarsening is *not* its mirror image. Redundancy/equality generally requires separate evidence and is handled at explicit recompression checkpoints:

coarsen only when redundancy is separately certified.

The familiar shorthand “coarsen when unnecessary; refine when necessary” should therefore be read as a design objective, not as a symmetric online mechanism.

5.3 Synergy and witness hypergraphs

Some hidden structures have no informative one-coordinate direction. A parity rule is the canonical example: each variable may have zero marginal predictive value while a higher-order combination is fully predictive. This phenomenon is standard in multivariate information theory and is naturally described using synergy/PID language [6].

Let V be the primitive candidate set. Define a witness hypergraph

$$\mathfrak{W} = (V, \mathcal{E}_W),$$

whose hyperedges $e \in \mathcal{E}_W$ are minimal consequential subsets under the chosen criterion. High-order antichains may be exponentially large, so witness search can be intrinsically combinatorial.

Established result / project re-derivation

Absence of low-order predictive signal does not imply absence of hidden structure. Parity gives an exact counterexample; this observation is established, not novel. The project-specific question is how a resource-bounded articulator reacts to that failure of greedy direction.

5.4 Blind Information Well

The frozen Blind Information Well milestone operationalizes articulation discovery within a declared finite grammar. Candidate articulations are subsets of grounded lag coordinates with learned conditional laws. The search follows:

1. begin from the null/coarsest articulation;
2. test low-cost directed refinements;
3. expand promising candidates;
4. reserve non-greedy exploration slots;
5. invoke synergy fallback when no marginal direction is informative;
6. select by validation evidence plus a structural complexity penalty;
7. respect an explicit search budget.

The conceptual output is a *discoverability profile*: structure may be statistically present yet expensive to locate. Metacognition should therefore represent not only uncertainty about the world, but uncertainty induced by exhausted search.

The Blind Information Well validates endogenous articulation selection *within a supplied primitive grammar*. On 200 blind wells from five generator families, the generous search achieved mean excess Bayes log loss 0.000729 bits/symbol, exact support recovery 97.5%, and mean support F1 0.9942, using 125.87/163 candidate evaluations on average. At a hard budget of 35 evaluations, recovery remained complete on several low-order families but degraded on arbitrary sparse-table and parity cases, exposing a resource/discoverability frontier. It does not establish arbitrary primitive-feature invention.

6 Persistent Entities and Transformation Invariance

6.1 The entity problem

A stream may contain repeated observations of the same latent entity under nuisance transformations. Let $x_i, x_j \in \mathbb{R}^d$ be two observations and let a finite transformation group/family G act on observations.

Hardly choosing one relative transformation too early is brittle. The robust formulation retains a distribution:

$$q_{ij}(g) = \frac{\exp[-\kappa_{\text{tr}} d_{\text{obs}}(x_j, g \cdot x_i)]}{\sum_{h \in G} \exp[-\kappa_{\text{tr}} d_{\text{obs}}(x_j, h \cdot x_i)]}, \quad g \in G.$$

Here d_{obs} is an observation-mismatch function and $\kappa_{\text{tr}} > 0$ is an inverse-temperature/concentration parameter. These symbols are intentionally distinct from the dynamic distortion $D_{\text{dyn}}^{(1)}$ and from streaming failure budgets.

6.2 Cycle consistency and discrete holonomy

For hard relative-transform representatives g_{ij} , a globally consistent assignment should satisfy

$$\Omega_G(i, j, k) = g_{ij} g_{jk} g_{ki} = e$$

on every consistent cycle. Nontrivial Ω_G is a discrete holonomy defect. In the soft setting, cycle consistency supplies a reliability score for transformation edges.

This creates an important separation:

$$\text{pairwise fit} \neq \text{globally synchronized transform structure.}$$

6.3 Global synchronization and gauge

Suppose there exist latent absolute transformations h_i such that

$$g_{ij} = h_j h_i^{-1}.$$

The h_i are identifiable only up to a common left/right gauge (depending on convention): applying the same global transform to all absolute states leaves every relative transform unchanged. Relative transformations are therefore gauge-invariant observables.

6.4 Selective nuisance subgroup

Maximal invariance is not automatically desirable: an overly large nuisance group can collapse genuinely distinct entities. Candidate subgroups $H \leq G$ are therefore scored by normalized evidence. A generic form is

$$\mathcal{L}(H) = \sum_{(i,j)} w_{ij} \log \left(\frac{1}{|H|} \sum_{g \in H} q_{ij}(g) \right),$$

where w_{ij} incorporates reliability/cycle consistency. The active subgroup is

$$\hat{H} = \arg \max_H \mathcal{L}(H),$$

and the evidence gap

$$\Delta_H = \mathcal{L}(\hat{H}) - \max_{H \neq \hat{H}} \mathcal{L}(H)$$

acts as a confidence margin.

6.5 Structural entity proposals and predictive veto

Under the selected subgroup, persistent H_0 partitions or other structural clusters can generate candidate entity identities. But structural persistence is only a proposal mechanism. A predictive common-refinement veto prevents a structurally attractive merge from being accepted if it destroys a consequential predictive distinction.

Thus entity identity is constrained by two independent kinds of evidence:

$$\boxed{\text{structural persistence} \quad + \quad \text{predictive compatibility.}}$$

6.6 Prototype refinement

When a structural partition is fragile, latent entity prototypes may be refined. Cortex v0.7 uses transformation-aligned sufficient statistics. For entity e , with aligned observations $\tilde{x}_t$, a simple prototype is

$$\mu_e = \frac{1}{N_e} \sum_{t:z_t=e} \tilde{x}_t.$$

The important architectural point is not this particular estimator; it is that prototype refinement is gated by evidence and participates in joint entity-invariance co-refinement.

Persistent entities, learned nuisance invariance/synchronization, joint entity-invariance co-refinement, and subgroup/gauge recovery are validated finite learning components. They are not currently general statistical consistency theorems.

7 Simultaneous Binding

7.1 Why frame-level assignment must be injective

If multiple detections appear simultaneously, independent nearest-prototype matching can assign two detections to the same entity. Cortex instead solves a joint assignment problem.

Let detections be $d_1, \dots, d_k$ and active entity hypotheses $e_1, \dots, e_n$. Define a cost

$$C_{ij} = \min_{g \in \hat{H}} D_w(d_i, g \cdot \mu_{e_j}),$$

where D_w is reliability-weighted error. Add spawn/dummy columns with cost c_{spawn} . The frame-level binding solves

$$\min_{\sigma} \sum_{i=1}^k C_{i, \sigma(i)}$$

subject to injectivity over existing entity columns.

This is a linear assignment problem. If no existing entity is sufficiently compatible, a new identity hypothesis is spawned.

7.2 Bootstrap guard

Before nuisance synchronization is confident, a large transformation family can make distinct entities spuriously similar. Cortex v0.7 therefore uses a conservative spawn gate during subgroup bootstrap. Once subgroup evidence is strong enough, normal matching under $\hat{H}$ resumes.

7.3 Resource boundary

Injectivity is treated as an invariant. If entity capacity is exhausted, the runtime raises an explicit capacity failure rather than silently merging simultaneous detections. This turns memory pressure into an observable resource failure rather than a hidden semantic error.

8 Residuals, Reliability, and Endogenous Noise Revision

8.1 Residual evidence

After alignment and binding, compare observation $\tilde{x}$ with entity prototype μ_e . Coordinate residuals are

$$r_{e,k} = |\tilde{x}_k - \mu_{e,k}|.$$

A residual above a threshold produces an anomaly indicator $a_{e,k} \in \{0, 1\}$. Cortex uses Beta/Bernoulli sufficient statistics

$$u_{e,k} \leftarrow u_{e,k} + a_{e,k}, \quad v_{e,k} \leftarrow v_{e,k} + 1 - a_{e,k},$$

with posterior anomaly probability

$$\hat{p}_{e,k} = \frac{u_{e,k}}{u_{e,k} + v_{e,k}}.$$

A reliability weight may be derived as

$$w_{e,k} = \text{clip}(1 - \hat{p}_{e,k}, w_{\min}, 1).$$

8.2 Three-way residual state

The system does not force every residual into “signal” or “noise.” Instead it can maintain:

$$\text{structure-supported}, \quad \text{unresolved}, \quad \text{noise-supported}.$$

This preserves epistemic uncertainty when evidence is insufficient.

An unexplained residual is not noise merely because the current model cannot explain it.

9 Relations, Events, and Causal Evidence

9.1 Relation/event posteriors

For a pair of persistent entities (a, b) , let $Y_t^{ab} \in \{0, 1\}$ denote an observed relation/event indicator when both entities are visible. A Beta-Bernoulli learner maintains

$$u_{ab} \leftarrow u_{ab} + Y_t^{ab}, \quad v_{ab} \leftarrow v_{ab} + 1 - Y_t^{ab},$$

with

$$\hat{p}_{ab} = \frac{u_{ab}}{u_{ab} + v_{ab}}.$$

After sufficient exposure, thresholds can classify the relation as active, absent, or unresolved. This is a finite validated mechanism, not a claim that every semantic relation is Bernoulli.

9.2 Intervention-sensitive causal contrast

Predictive association is not promoted to causation. For ordered pair $a \rightarrow b$, maintain outcome posteriors under intervention on a and matched non-intervention control:

$$p_{\text{do}}^{a \rightarrow b} = \mathbb{E}[Y_b \mid \text{do}(a)], \quad p_{\text{ctrl}}^{a \rightarrow b} = \mathbb{E}[Y_b \mid \text{matched control}].$$

The finite implementation uses posterior means and contrast

$$\Delta_{a \rightarrow b} = p_{\text{do}}^{a \rightarrow b} - p_{\text{ctrl}}^{a \rightarrow b}.$$

Large positive contrast supports a causal hypothesis; small contrast supports a null; insufficient exposure remains unresolved.

The causal branch is validated only in a narrow intervention protocol. It is not a general solution to causal discovery under hidden confounding, selection bias, arbitrary interventions, or continuous treatments.

10 Streaming Articulation

10.1 Finite online state

For an unbounded observation stream

$$X_1, X_2, \dots \in \mathcal{X},$$

Lambda operates causally through a finite articulated state

$$S_{t+1} = U(S_t, X_{t+1}).$$

The state contains learned quotient/memory, sufficient statistics, uncertainty, and unresolved witnesses required for future processing. It should not retain the entire raw history unless the application explicitly chooses to.

The architecture has two timescales:

$$S_{t+1} = U_{\text{maintain}}(S_t, X_{t+1})$$

when the retained articulation remains supported, and

$$S_{t+1} = U_{\text{escalate}}(S_t, X_{t+1}; B_t)$$

when consequential evidence triggers a bounded structural search budget B_t .

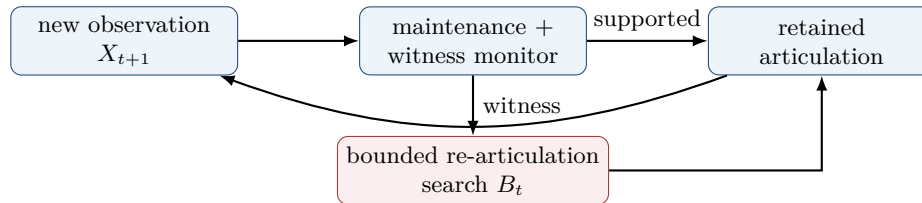


Figure 2: Fast maintenance path and slow witness-triggered escalation.

10.2 Piecewise-stationary theorem interface

The frozen restricted streaming theorem applies after an upstream articulation has produced a finite, fully visible state process $Z_t \in \mathbf{X}$ satisfying a Markov closure within each stationary regime:

$$\mathbb{P}(Z_{t+1} \mid Z_{1:t}) = \mathbb{P}(Z_{t+1} \mid Z_t).$$

Let the regime kernels be

$$K^{(0)}, K^{(1)}, \dots, \quad 0 = \tau_0 < \tau_1 < \tau_2 < \dots.$$

Each regime has a canonical memory $\mathcal{P}_\star^{(q)}$.

10.3 Coverage and recovery horizons

Suppose regime q has an L_q -step covering certificate

$$\eta_q = \min_{x,i} (K^{(q)})^{L_q}(x, i) > 0.$$

For n required departures per state and coverage-failure budget $\delta \in (0, 1)$, define

$$T_{\text{cov}}(n, \delta; L_q, \eta_q) = 1 + L_q \left\lceil \frac{n + c + \sqrt{c^2 + 2nc}}{\eta_q} \right\rceil, \quad c = \log \frac{m}{\delta}.$$

Then the probability of failing to obtain n departures from every state by that horizon is at most δ . This is an inherited finite-time coverage lemma: the certificate η_q lower-bounds the chance of reaching each state in an L_q -step block, and a binomial-tail bound plus a union bound over m states yields the displayed sufficient horizon.

If exact recovery requires a positive memory-path margin lower bound

$$0 < \underline{\Delta}_M^{(q)} \leq \Delta_{M,\text{true}}^{(q)},$$

set

$$n_{\text{mem}}(\zeta; q) = \left\lceil \frac{8m^2}{(\underline{\Delta}_M^{(q)})^2} \log \frac{4m^2}{\zeta} \right\rceil$$

and

$$T_{\text{rec}}^{(q)}(\zeta) = T_{\text{cov}}\left(n_{\text{mem}}(\zeta; q), \frac{\zeta}{2}; L_q, \eta_q\right).$$

10.4 Anytime monitoring

Lifetime false-refinement control uses summable error schedules. For example,

$$\beta_k = \frac{6\alpha_{\text{mon}}}{\pi^2 k^2}, \quad \gamma_k = \frac{6\alpha_{\text{cov}}}{\pi^2 k^2}.$$

For a promised detectable instability margin $\Delta_d > 0$, define

$$n_k = \left\lceil \frac{9m^2}{2\Delta_d^2} \log \frac{2m^2}{\beta_k} \right\rceil,$$

then

$$T_{\text{mon}}^{(q)}(k) = T_{\text{cov}}(n_k, \gamma_k; L_q, \eta_q).$$

The monitor index is global and never reset to reclaim error budget.

10.5 Structural invalidation versus kernel change

A kernel can change without invalidating the retained articulation. Define the retained exact target recursively. If

$$D_{\text{dyn}}^{(1)}(\overline{\mathcal{P}}^{(q-1)}; K^{(q)}) = 0,$$

keep the old target. If

$$D_{\text{dyn}}^{(1)}(\overline{\mathcal{P}}^{(q-1)}; K^{(q)}) \geq \Delta_d,$$

the change is a witnessable structural invalidation and the system recovers $\mathcal{P}_\star^{(q)}$.

Hence

$$J_{\text{art}}(T) = J_{\text{str}}(T) + J_{\text{cmp}}(T)$$

counts articulation events, while $J_{\text{ker}}(T)$ counts kernel changes. In general these are not equal.

10.6 What is proved here versus inherited

The restricted streaming result is a *composition theorem*. Its primitive finite-sample ingredients are taken as proved lemmas from the frozen consolidation gaps: (i) raw-time coverage from L_q, η_q , (ii) finite-sample exact recovery under a positive memory-path margin, and (iii) time-uniform monitoring/false-alarm control on homogeneous blocks. This paper reproduces their interfaces and constants and gives the composition argument below. It does not re-prove every concentration constant from first principles.

The composition is checkable from four steps:

1. allocate summable lifetime budgets to monitoring, coverage, and recovery so that the union of all corresponding failure events has probability at most α_{tot} ;
2. convert required per-state sample counts to raw time using T_{cov} ;
3. impose deterministic dwell envelopes large enough to finish detection plus exact recovery before the next witnessable regime change;
4. outside those adaptation intervals, use minimality of the current canonical stable partition to conclude that the retained target is dynamically stable (possibly over-refined after pure simplification).

10.7 Restricted streaming theorem

The theorem allocates recovery budgets

$$\zeta_s = \frac{6\alpha_{\text{rec}}}{\pi^2(s+1)^2}$$

and a deterministic adaptation envelope

$$\overline{A}_s(t) = T_{\text{mon}}^{(q_s)}(t+1) + T_{\text{mon}}^{(q_s)}(t+2) + T_{\text{rec}}^{(q_s, s)}.$$

With sufficient dwell time between witnessable structural invalidations, the theorem guarantees, with total failure probability bounded by

$$\alpha_{\text{tot}} = \alpha_{\text{mon}} + \alpha_{\text{cov}} + \alpha_{\text{rec}},$$

that initial exact memory is recovered, false structural refinements are lifetime-controlled on theorem-eligible homogeneous blocks, witnessable invalidations are detected, and the new exact canonical memory is reintegrated within the stated envelope.

This is a restricted composition theorem. It assumes an upstream finite articulated Markov carrier, valid margins/certificates, homogeneous monitoring blocks, and sufficient dwell. It does not prove that arbitrary raw sensory streams are automatically converted into such a carrier.

Composition proof sketch. Let E_{mon} , E_{cov} , and E_{rec} denote the unions of all monitoring, raw-time coverage, and exact-recovery failure events over the lifetime schedule. Because

$$\sum_{k \geq 1} \frac{6}{\pi^2 k^2} = 1, \quad \sum_{s \geq 0} \frac{6}{\pi^2 (s+1)^2} = 1,$$

the schedules imply

$$\mathbb{P}(E_{\text{mon}}) \leq \alpha_{\text{mon}}, \quad \mathbb{P}(E_{\text{cov}}) \leq \alpha_{\text{cov}}, \quad \mathbb{P}(E_{\text{rec}}) \leq \alpha_{\text{rec}}.$$

Hence by the union bound the structural good event

$$E_{\text{good}} = (E_{\text{mon}} \cup E_{\text{cov}} \cup E_{\text{rec}})^c$$

has probability at least $1 - \alpha_{\text{tot}}$.

On E_{good} , initial exact recovery finishes by $T_{\text{rec}}^{(0)}$. A theorem-eligible homogeneous monitoring block that contains no structural invalidation cannot create a false certified split. If a new regime creates distortion at least Δ_d , the monitoring lemma detects the positive witness after at most the two conservative monitor-block horizons appearing in $\overline{A}_s$; fresh post-change data then satisfy the exact-recovery lemma by $T_{\text{rec}}^{(q_s, s)}$. The dwell assumption guarantees that this detection/recovery episode completes before the next kernel change. Induction over witnessable structural events therefore yields the retained-target claim for all times outside the union of adaptation intervals. The bound on $H_{\text{str}}(T)$ is the sum of their deterministic lengths. After a non-invalidating kernel change, the old retained partition remains dynamically stable, so by coarsest-stable-partition minimality it is a stable refinement of the new canonical minimum. This also explains why exact automatic coarsening is not promised. $\square$

10.8 Regret bound

For a task-relative consequence target $\mathcal{Y}$, suppose per-symbol loss lies in $[0, L_{\text{max}}]$, and when the articulation is stable the maintenance predictor has excess expected loss at most $\varepsilon_{\text{pred}}$. Then on the structural good event,

$$\boxed{\text{Reg}_{\mathcal{Y}}(T) \leq \varepsilon_{\text{pred}} T + L_{\text{max}} H_{\text{str}}(T).}$$

Thus

$$\frac{\text{Reg}_{\mathcal{Y}}(T)}{T} \leq \varepsilon_{\text{pred}} + L_{\text{max}} \frac{H_{\text{str}}(T)}{T}.$$

Under uniform margins and

$$J_{\text{str}}(T) \log T = o(T),$$

we obtain

$$\limsup_{T \rightarrow \infty} \frac{\text{Reg}_{\mathcal{Y}}(T)}{T} \leq \varepsilon_{\text{pred}}.$$

A separate maintenance theorem would be needed to set $\varepsilon_{\text{pred}} = 0$.

11 Computation and Amortization

11.1 Maintenance plus discovery

Let c_{maint} bound ordinary per-symbol maintenance work and let S_a be the structural-search cost for articulation event a . Then

$$C(T) \leq c_{\text{maint}}T + \sum_{a=0}^{J_{\text{art}}(T)} S_a.$$

If the chosen finite carrier/grammar provides a finite event-wise bound $S_a \leq S_{\text{max}}$, then

$$C_{\text{search}}(T) \leq S_{\text{max}}(J_{\text{art}}(T) + 1).$$

Hence

$$J_{\text{art}}(T) = o(T) \implies \frac{C_{\text{search}}(T)}{T} \rightarrow 0.$$

This formalizes *amortized articulation*: discovery may be expensive, but its overhead per observation vanishes when structural reopenings are sublinear.

11.2 Entropy is not the controller

A zero-entropy source can have arbitrarily many predictive states. Take a binary primitive word of period p and choose the phase uniformly. The entropy rate is

$$h_{\mu} = 0,$$

yet each of the p phases has a different future, so

$$|Q^*| = p.$$

Therefore

$$h_{\mu} = 0 \not\Rightarrow |Q^*| \text{ bounded.}$$

The relevant resource variables include quotient size, memory entropy, predictive separation margins, drift, witness frequency, and unresolved synergy-search cost.

11.3 What amortization does not solve

Cheap recognition after learning does not imply cheap discovery. Difficult cases remain:

- high-order synergy search;
- exponentially large witness antichains;
- large transformation grammars;
- large entity-assignment spaces;
- unbounded predictive-state complexity;
- rapid regime switching;
- vanishing predictive margins.

12 Cortex v0.7: Dependency-Directed Execution

12.1 Operational state

Cortex v0.7 is a contract-faithful integrated runtime for the validated entity/invariance/binding/noise/relation/causal ideas. Its online state contains:

- persistent entity prototypes;
- coordinate reliability and residual/noise statistics;
- per-entity relative-transform evidence;
- nuisance-subgroup evidence and confidence margin;
- simultaneous frame bindings;
- pairwise relation/event posteriors;
- ordered intervention/control causal contrasts.

12.2 Local statistics, local re-derivation

Cortex separates *sufficient-statistic update* from *articulated-state re-derivation*. For every observation:

1. bind detections injectively to entity hypotheses;
2. update all sufficient statistics touched by the observation;
3. re-derive only articulated cells whose statistics changed;
4. propagate invalidation only when a re-derived articulated value actually changes;
5. preserve all unaffected articulation in memory.

Write Δ_t for the local statistical change at time t , and let V_{art} be the active articulation dependency graph. The realized recomputation closure is

$$\text{Dep}^*(\Delta_t) \subseteq V_{\text{art}}.$$

Define the locality ratio

$$\rho_t = \frac{|\text{Dep}^*(\Delta_t)|}{|V_{\text{art}}|}.$$

The execution hypothesis is that, for locally changing worlds, ρ_t can remain small even when the represented world grows.

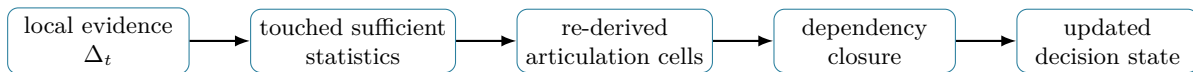


Figure 3: Cortex local re-derivation rule. Learning continues locally; only consequential state changes propagate.

12.3 Synchronization hardening

Relative-transform evidence is accepted only after identity persistence exists. A first observation of a freshly spawned entity cannot be compared with itself to create a false zero-transform witness. Evidence is normalized per established entity, then combined by robust consensus so that a fragmented/high-count track cannot dominate by volume alone.

12.4 Entity-invariance co-refinement guard

Before the nuisance subgroup is established, an overly permissive transformation family can make distinct entities appear equivalent. Cortex therefore uses a conservative matching/spawn threshold during synchronization bootstrap

and relaxes it only after subgroup confidence is adequate.

This is an example of a broader Sigma-Lambda pattern:

when uncertainty in one articulation layer can corrupt another, delay irreversible merging.

12.5 Capacity as an explicit resource boundary

If simultaneous binding requires a new entity but the declared entity budget is exhausted, Cortex raises an explicit capacity failure. It does not silently violate injectivity. More generally, resource failures are intended to be represented as metacognitive state rather than converted into hidden semantic errors.

12.6 Frozen v0.7 evidence and internal baseline

The v0.7 empirical evidence is a controlled synthetic contract suite, not a real-world validation campaign. The principal computational baseline is *eager re-derivation*: the same learned sufficient statistics, semantics, and update rules are used, but all articulated cells are recomputed after every observation. The dependency runtime is therefore compared against an internal exact baseline rather than against a different model.

The final moderate-noise runtime suite used persistent-entity worlds with $n \in \{8, 12, 16\}$, 4–6 simultaneous visible entities, 110–150 observations per run, additive Gaussian observation noise with standard deviation 0.06, independent coordinate corruption probability 0.035, and both stable and local-drift regimes. Across eight such cases:

- eager/dependency articulated-state agreement: **100%**;
- mean identity accuracy: **98.23%** (minimum 97.27%);
- nuisance-subgroup recovery: **100%**;
- resolved relation accuracy: **100%**, with 98.71% mean resolved coverage;
- mean articulated cells reopened: **19.56%**; at $n = 16$, **12.69%**;
- mean end-to-end speedup over eager re-derivation: **1.073×**; one small case was slower ($0.917\times$).

The full freeze gate comprised nine behavioral contract groups and 2,200 sequential eager/dependency comparisons with zero binding or articulated-state mismatches. A harder intermittent-visibility campaign achieved 99.894% mean identity accuracy (99.762% minimum) after the entity-invariance bootstrap repair. A separate severe-corruption sweep showed the opposite frontier: as coordinate corruption rose from 0.01 to 0.20, the number of entity hypotheses for eight true entities rose from 11 to 115 while the residual state moved predominantly to “unresolved”. This is evidence of conservative uncertainty preservation but also of a real memory-growth failure mode.

These numbers support *contract fidelity and locality inside the controlled benchmark family*. They do not establish external validity, universal speed advantage, or general statistical consistency. Neural and external-stream comparisons are reported separately and are not used as proof of the formal claims in this document.

Cortex v0.7 supports the operational statement: **continuous local learning does not require continuous global re-articulation**. This is validated for the current finite learning stack, not proved for arbitrary architectures or environments.

13 Status Ledger: What Is Proved, Validated, Conditional, or Open

Layer	Status	Interpretation
Substrate inexhaustibility	Scope guard	Ontological anti-totalization principle; not required for most finite proofs.
Coherent articulation maps	Formal axiom	Defines admissible resolution/refinement structure.
Continuation-law existence	Formal axiom	Consequence-relative stochastic interface.
Predictive equivalence	Exact	Equality of continuation laws.
Predictive quotient minimality	Established / re-derived exact	Coarsest sufficient predictive factor under stated conditions.
Projectability across resolutions	Exact criterion	Quotient tower exists only when fine classes do not straddle coarse quotient states.
Dynamic sufficiency	Established / specialized exact finite	Lumpability/stability of partitions.
Minimal stable memory	Project finite specialization	Unique coarsest dynamically stable refinement of a supplied scaffold.
Operational topology / ultrametric	Exact conditional constructions	Follow from reachability or nested refinement assumptions.
Fisher geometry	Conditional	Requires a regular parametric continuation-law family.
Witness/synergy formalism	Established phenomenon + project search formalism	Discovery search can remain combinatorial.
Blind articulation discovery	Validated finite benchmark	Endogenous within a declared primitive grammar.
Entities + invariance	Validated finite learner	Controlled identity/synchronization campaigns; no general consistency theorem.
Binding + residual/noise	Validated finite learner	Explicit injectivity and three-way residual state.
Relations/events	Validated finite learner	Controlled pairwise relation protocol.
Causal contrast	Validated narrow protocol	Intervention-sensitive evidence; not general causal discovery.
Restricted streaming theorem	Project restricted composition theorem	Requires finite visible Markov carrier, margins, coverage certificates, homogeneous monitor blocks, dwell.
Cortex v0.7 dependency execution	Validated execution milestone	Eager/local equivalence on frozen contract suite; locality is empirical.
Infinite-state minimal memory	Open	Finite termination does not directly generalize.
Online quotient consistency	Open	Needed for general unbounded streams.
Discovery-horizon theorem	Open	Needed to derive finite discovery time from source assumptions.
General approximate quotient theory	Open	Needed for irreducibly infinite predictive complexity.
General feature-grammar invention	Open	Current endogenous discovery assumes grounded primitives/grammar.

14 Known Failure Frontiers

14.1 Search complexity

The theory deliberately keeps discovery cost visible. A candidate space can be too large even when the correct articulation is simple. Synergy is the cleanest example: no marginal witness may indicate the high-order structure.

14.2 Equality is harder to certify than inequality

Positive difference can be witnessed. Exact equality often lies on a statistical boundary and may not admit finite-time certification without extra assumptions. This asymmetry explains why the automatic branch refines on positive witnesses but treats coarsening more cautiously.

14.3 Pure simplification

A new regime can make an old distinction unnecessary while leaving the old finer partition dynamically stable. Then no positive witness forces forgetting. Exact automatic tracking can remain safely over-refined until an explicit recompression checkpoint certifies redundancy.

14.4 Identity fragmentation under strong corruption

Conservative identity handling avoids false merges, but under severe corruption it can proliferate temporary identity hypotheses. This is a genuine resource/corrigibility tradeoff:

$$\text{avoid false merge} \iff \text{accept temporary hypothesis growth.}$$

A future certified hypothesis reconciliation/pruning layer may address this frontier.

14.5 External validity

Most formal learning evidence remains synthetic or controlled. The current roadmap therefore prioritizes task-free continual graph/interaction streams not designed specifically for Cortex.

15 Interpretive Summary

The modern Sigma-Lambda program can be summarized in five mathematical commitments.

- 1. Consequence-relative minimality.** Representations should preserve distinctions only when those distinctions matter to the chosen continuation/consequence target.
- 2. Asymmetric structural revision.** Refinement is automatically witness-driven: positive evidence that a merge is consequential triggers a split. Coarsening is not the mirror operation; it occurs only when redundancy is separately certified, generally at an explicit recompression checkpoint.
- 3. Resource-relative approximation.** When exact memory is too expensive or infinite, choose a coherent approximate articulation on a risk-cost frontier rather than allowing uncontrolled memory growth.
- 4. Structural uncertainty must remain visible.** Unresolved residuals, exhausted search budgets, subgroup ambiguity, and capacity exhaustion should not be silently converted into false certainty.
- 5. Continuous learning need not imply global recomputation.** Cortex operationalizes the idea that new evidence should update local sufficient statistics and only propagate as far as the learned consequences actually change.

These commitments yield the project-wide computational picture below as an *architectural objective*. No current theorem proves the full observation-to-causal-evidence chain end to end, and no general uncertainty-propagation

theorem connects all modules.

observe $\rightarrow$ articulate $\rightarrow$ quotient $\rightarrow$ stabilize $\rightarrow$ maintain $\rightarrow$ reopen on witness.

16 Roadmap

The next mathematically and scientifically important developments are:

1. **External task-free continual articulation.** Test whether witness-driven stabilization and local revision survive on real temporal graph/interaction streams with no supplied task boundaries.
2. **Online quotient consistency.** Prove convergence of an online articulation estimator under mixing/separation assumptions.
3. **Approximate predictive quotient theory.** Develop coherent resource-bounded quantizers for sources with infinite exact predictive complexity.
4. **Certified hypothesis reconciliation.** Reduce identity-hypothesis proliferation without introducing unsafe merges.
5. **Feature-grammar expansion.** Allow the discovery layer to propose temporal transforms, compositions, latent states, and multiscale components rather than only selecting within a fixed primitive grammar.
6. **Hybrid neural compilation.** Use Cortex for structural discovery/revision and compile stable high-frequency articulations into fast learned recognizers where appropriate.

A Notation Reference

Symbol	Meaning
Σ	Substrate-excess; not itself a finite learned state.
Λ	Articulation/representational process over grounded observations.
X_t	Grounded observation at time t .
A_r	Articulation map at resolution r .
$\pi_{s \rightarrow r}$	Reduction map from finer resolution s to coarser resolution r .
$\mathcal{L}(U \mid V = v)$	Conditional probability law/distribution of U given $V = v$.
$\Psi_r(x)$	Continuation law of articulated state x .
$x \equiv_r y$	Predictive equivalence (identical continuation laws).
Q_r	Predictive quotient at resolution r .
$K, K^{(q)}$	Finite-state Markov transition kernel; $K^{(q)}$ is regime q .
$\Pi, \mathcal{P}$	Partition/articulation of a finite carrier.
$\mathcal{P}_\star$	Coarsest dynamically stable refinement of a supplied scaffold.
$\mathcal{M}_\star$	Minimal stable-memory quotient $\mathbf{X}/\mathcal{P}_\star$.
$D_{\text{dyn}}^{(1)}$	One-step dynamic stability distortion.
d_{obs}	Observation mismatch used in transformation evidence.
$C^\star(\varepsilon)$	Minimum representation cost achieving predictive risk at most ε .
$\mathfrak{W} = (V, \mathcal{E}_W)$	Witness hypergraph over primitive candidates.
G	Candidate finite transformation group/family.
$H \leq G$	Candidate nuisance subgroup.
$q_{ij}(g)$	Soft relative-transformation evidence.
κ_{tr}	Inverse-temperature/concentration in the transformation softmax.

Symbol	Meaning
Ω_G	Transformation-cycle holonomy defect.
$\hat{H}$	Selected nuisance subgroup.
Δ_H	Subgroup evidence margin.
$u_{e,k}, v_{e,k}$	Beta posterior pseudo-counts for coordinate-residual anomaly probability.
u_{ab}, v_{ab}	Beta posterior pseudo-counts for pairwise relation probability.
S_t	Finite online articulated state.
δ	Generic finite-horizon coverage failure budget in T_{cov} .
$\alpha_{\text{mon}}, \alpha_{\text{cov}}, \alpha_{\text{rec}}$	Lifetime monitoring, coverage, and recovery failure budgets.
$\beta_k, \gamma_k, \zeta_s$	Summable per-block/per-recovery allocations derived from the lifetime budgets.
Δ_d	Promised detectable structural-invalidation margin.
$J_{\text{ker}}, J_{\text{str}}, J_{\text{art}}$	Kernel changes, witnessable structural invalidations, and articulation events.
$H_{\text{str}}(T)$	Total deterministic upper bound on exact structural adaptation time up to horizon T .
ρ_t	Cortex realized dependency/locality ratio.

B Compact Proof Sketch: Why Stable Refinement Terminates in the Finite Case

Let $\mathbf{X}$ contain $m < \infty$ states. Every strict refinement increases the number of partition blocks by at least one, and the number of blocks is at most m . Therefore a synchronous refinement sequence can contain at most $m - |\mathcal{P}_0|$ strict refinement steps. When refinement stops, no block contains a pair with different aggregated continuation laws, so the terminal partition is dynamically stable. Any dynamically stable refinement of $\mathcal{P}_0$ must separate every pair separated by the refinement operator; therefore the terminal partition is coarser than or equal to every other dynamically stable refinement. This gives uniqueness of the coarsest stable refinement.

C Proof Architecture of the Restricted Streaming Composition

The restricted theorem composes three lifetime families of events. Let F_k^{mon} be the false-monitor event on monitoring block k , F_k^{cov} the corresponding raw-time coverage failure, and F_s^{rec} the exact-recovery failure on recovery episode s . The schedules

$$\beta_k = \frac{6\alpha_{\text{mon}}}{\pi^2 k^2}, \quad \gamma_k = \frac{6\alpha_{\text{cov}}}{\pi^2 k^2}, \quad \zeta_s = \frac{6\alpha_{\text{rec}}}{\pi^2 (s+1)^2}$$

obey

$$\sum_k \beta_k = \alpha_{\text{mon}}, \quad \sum_k \gamma_k = \alpha_{\text{cov}}, \quad \sum_s \zeta_s = \alpha_{\text{rec}}.$$

If the primitive lemmas guarantee $\Pr(F_k^{\text{mon}}) \leq \beta_k$, $\Pr(F_k^{\text{cov}}) \leq \gamma_k$, and $\Pr(F_s^{\text{rec}}) \leq \zeta_s$, then Boole's inequality gives

$$\Pr\left(\bigcup_k F_k^{\text{mon}} \cup \bigcup_k F_k^{\text{cov}} \cup \bigcup_s F_s^{\text{rec}}\right) \leq \alpha_{\text{mon}} + \alpha_{\text{cov}} + \alpha_{\text{rec}}.$$

On the complementary good event, all completed homogeneous stable blocks are free of false certified refinement, every witnessable invalidation receives enough covered data to be detected, and every ensuing recovery episode receives enough fresh post-change departures to recover the finite canonical partition. The deterministic dwell envelope ensures these episodes do not overlap the next kernel change. Induction over structural invalidations yields the retained-target claim. The regret corollary then follows by splitting time into adaptation intervals (worst-case

loss at most $L_{\max}$) and stable intervals (excess expected loss at most $\varepsilon_{\text{pred}}$).

What remains external to this composition proof are the primitive concentration inequalities that yield the displayed constants in T_{cov} , n_{mem} , and n_k . Their assumptions are exposed in Section 10 rather than hidden behind the theorem label.

D Internal Technical Provenance

This whitepaper consolidates the following frozen/current project artifacts:

- *Sigma–Lambda Formal Consolidation Audit*, 2026-09-13.
- *Corrected Restricted Streaming Articulation Theorem v0.2*, 2026-09-13.
- *Blind Information Well / Endogenous Articulation Discovery v0.1 Freeze Record*, 2026-09-11.
- *Cortex v0.7 Integrated Articulation Runtime - Architecture and Freeze Record*, 2026-09-14.

The exact historical *Consolidated Implementation v0.3* source archive was not recoverable during the Cortex v0.7 freeze. Cortex v0.7 is therefore a contract-faithful frozen runtime rather than a byte-for-byte claim of identity with that historical source.

E Selected Mathematical Context and Provenance

The present program deliberately combines established mathematical traditions. Predictive equivalence and minimal sufficient predictive states are directly connected to computational mechanics/causal states [2]. Finite dynamic stability is classical Markov lumpability/state aggregation [1], with the coarsest-refinement algorithmic viewpoint closely related to partition-refinement work such as Paige–Tarjan [3]. Resource–risk compression is structurally related to Shannon rate–distortion and the information bottleneck [4, 5]. Pure synergy and the failure of marginal signals are standard multivariate-information phenomena [6]. The entity layer borrows group-synchronization ideas [8], while Cortex’s execution layer is conceptually adjacent to the self-adjusting-computation tradition.

Accordingly, the research claim should not be read as “these component theorems are new”. The project-level questions are whether these ingredients can be composed coherently under explicit resource and uncertainty semantics; whether the negative results (non-projectability, equality/simplification asymmetry, entropy insufficiency, discoverability frontiers) constrain that composition in useful ways; and whether dependency-directed continual articulation remains empirically useful outside controlled worlds.

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